

Meihui (Lily) Wu

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EDUCATION

University of Southern California

Los Angeles, CA

B.S. Applied & Computational Mathematics; B.A. Cognitive Science

Expected May 2027

- **GPA: 3.87** | **Selected Coursework:** Data Science in Brain & Cognitive Science, Data Science with Python, Probability Theory & Mathematical Statistics, Neural Network Models of Social and Cognitive Processes, Behavioral Neuroscience

RESEARCH INTERESTS

Multimodal social interaction; computational modeling of conversation dynamics and social connection; affective computing; computational mental health; human-AI interaction

PUBLICATIONS & PREPRINTS

Zhou, C., **Wu, M.-H.**, Xiang, Y.-M., & Itti, L. (2026). *Cross-Modal Benchmarking of Acoustic Prosody and Ventral Striatal BOLD for Depression-Related Anhedonia Classification: A Pre-Registered Study with the ClinicalWhisper Pipeline*. bioRxiv preprint. [doi:10.64898/2026.06.08.728970](https://doi.org/10.64898/2026.06.08.728970)

CONFERENCE & RESEARCH PRESENTATIONS

Wu, M., Lu, C., Grady, S. M., & Baek, E. C. (2026). *Linguistic Alignment and Conversation Dynamics as Predictors of Social Connection in Dyadic Co-viewing*. Poster presented at the Social & Affective Neuroscience Society Annual Conference.

Zhou, C., **Wu, M.-H.**, Xiang, Y.-M., & Itti, L. (2026). *ClinicalWhisper: Cross-Modal Benchmarking of Acoustic Prosody and Ventral Striatal BOLD for Depression-Related Anhedonia Classification*. Poster presented at an undergraduate research symposium, University of Southern California.

HONORS & RESEARCH FUNDING

- USC Provost's Undergraduate Research Fellowship (PURF), Summer 2025 and Summer 2026
- Cathay Bank Scholarship, 2025 (one of six Southern California recipients)
- Lenore B. Kreiger Endowed Scholarship for Mathematics, 2025; Syllas and Rose Marx Meyer Scholarship, 2024

TECHNICAL & RESEARCH SKILLS

Programming & ML: Python (pandas, NumPy, scikit-learn, PyTorch), MATLAB, C++; transformers, model evaluation, cross-validation

Multimodal Behavior: SBERT/RobERTa, WhisperX, OpenSMILE, Parselmouth, OpenCV/MediaPipe; language, prosody, pose/gesture

Neurotechnology & Sensing: fNIRS, fMRI, EEG, Neuropixels/Kilosort4; signal processing, sensor integration, BLE

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Jan. 2025 – Present

USC Social Connection Lab (PI: Elisa Baek)

Los Angeles, CA

- Integrated 39 linguistic, acoustic, and visual features into an end-to-end Python pipeline spanning 24 dyads and 46 sessions; collect Aurora fNIRS and behavioral data.
- Quantified transcript semantics with SBERT and sentiment with RoBERTa; operationalized 10 turn-level interaction dimensions across 5,425 turns from 23 dyads using a few-shot LLM calibrated against researcher judgments on a held-out dyad.
- Generated synchronized prosodic and pose/gesture features by aligning speaker-corrected transcripts with WhisperX timestamps and processing audio/video with Parselmouth, librosa, OpenCV, and MediaPipe.
- Identified fine-grained turn exchange as the strongest correlate of partner evaluations and replicated turn-taking effects in 1,603 CANDOR dyads; assessed robustness with pseudo-dyad/time-shift nulls, reliability analysis, FDR correction, and nested leave-one-pair-out prediction.

RESEARCH EXPERIENCE (CONTINUED)

Undergraduate Research Assistant

Sep. 2024 – Present

USC Neural Modeling and Interface Lab (PI: Dong Song)

Los Angeles, CA

- Characterize memory-related CA1 activity through spike sorting and quantitative analysis for hippocampal circuit models.
- Converted raw electrophysiological recordings into Kilosort4-compatible formats using custom Python and MATLAB scripts, enabling a more efficient and reproducible spike-sorting workflow.
- Link behavioral and electrophysiological data while supporting high-density Neuropixels recordings during transcranial magnetic stimulation in Barnes Maze experiments.

Undergraduate Research Assistant

Jun. 2025 – Aug. 2025

USC Neural Vitality Lab

Los Angeles, CA

- Completed preprocessing and quality control for 496 BOLD scans, assessed 220 field maps for alignment, and reviewed 37 T1 images for tissue segmentation.
- Resolved signal dropout and misregistration artifacts affecting cross-site reproducibility.
- Scaled neuroimaging preprocessing and quality control through parallel HPC workflows while managing large-scale datasets.

Project Lead & Media Assistant

Sep. 2024 – Present

USC NeuroTech Club

Los Angeles, CA

- Achieved 75% accuracy in proficiency classification with a Python random forest pipeline for an interdisciplinary eye-tracking study of bilingual code-switching, covering preprocessing, feature engineering, and model evaluation.
- Implement an error-related potential BCI paradigm with the DSI-7 EEG system, including cap setup, impedance checks, artifact reduction, and early-stage signal inspection.
- Presented project findings to 400+ attendees at the 2025 California Neurotechnology Conference; coordinated milestones and delegated team deliverables.

SELECTED RESEARCH & ENGINEERING PROJECTS

ClinicalWhisper: Cross-Modal Anhedonia Classification

Python, Speech ML, Computational Psychiatry

- Benchmarked logistic regression, random forest, and gradient-boosted trees under preregistered stratified 5-fold cross-validation using 88 eGeMAPS features extracted from DAIC-WOZ interviews ($n = 142$) after Whisper Large-v3 transcription and pyannote diarization.
- Achieved AUC-ROC = 0.63 with the best voice model and benchmarked against ventral striatal BOLD ($n = 272$); SHAP analysis ranked pitch variability among the strongest predictors.

NeuroFoot: Smart Plantar-Pressure Monitoring System

Embedded Sensing, BLE, FMEA, Verification

- Engineered a 16-zone FSR sensing architecture from clinical requirements using a CD74HC4067 multiplexer and Arduino Nano 33, sampling pressure at 50 Hz and transmitting graded OK/WARN/CRITICAL alerts over BLE 5.0 to an iOS heatmap.
- Verified 9 of 10 components and achieved sub-200 ms alert latency through breadboard and integrated walk testing; converted ISO 14971 FMEA risks into design controls and defined a Class II 510(k) path against SmartStep (K060150).

KIN: Digital Care Coordination Platform

UX Research Lead, Figma

- Defined a dual-persona framework for patients and remote caregivers through competitive analysis across seven dimensions, journey mapping, and usability testing.
- Produced at least three design iterations per interface and evidence-based stakeholder recommendations from 5-second and task-based prototype tests.

ADDITIONAL EXPERIENCE

Communication Assistant

May 2025 – Present

USC Orientation Program

Los Angeles, CA

- Delivered usability-focused guides and event communications for students and guests across 10+ events, informed by survey analysis of 30+ Orientation Leaders.